

Manual

Processing Alternative / Parallel Sequences BDE-APF 8.2

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1 Overview: Processing Alternative / Parallel Sequences

Features

- Processing of alternative or parallel sequences
 - Supplementary function for processing alternative or parallel sequences in HYDRA
 - Configuration option used to activate sequence-based functions in HYDRA as part of customizing.
 - Management of standard sequence and alternative or parallel sequences
 - Option of transferring alternative sequences (customer-specific interface)
 - Tabular sequence display with option to replace sequences from alternative sequences in HYDRA to short-term production control (uploading the sequence replacement to SAP requires that a customer-specific interface has been realized)
 - Maintenance or processing dialog used to display or create order sequences
 - Consideration of the smallest quantity of a parallel sequence if validation checking for send-ahead quantities is enabled.
 - Transfer of parallel sequences from SAP PP via the PP-PDC interface
 - Consideration of sequences for time ticket-based uploads via PP-PDC interface to SAP PP.

Please note: Adding and coordinating the specific requirements and implementing them are considered a customized HYDRA service (a service subject to an added charge).

2 Edit Order Sequences

Overview

HYDRA menu	Order management → Order management → Edit order sequences
FEDRA menu	Detailed Scheduling → Order management → Edit order sequences
Transaction code	edseq
Function authorization	edseq

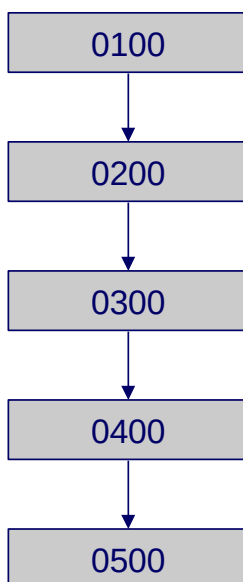
Purpose

If you want to group operations of an order, you can create sequences of operations in an order. In production, you use these sequences to specify the processing of operations. Within a sequence, the operations are processed one after the other according to the specified order. You can link several sequences. Using these links, you can integrate network-type structures.

You can also use parallel or alternative order sequences. The following sequence types are supported:

Standard sequence

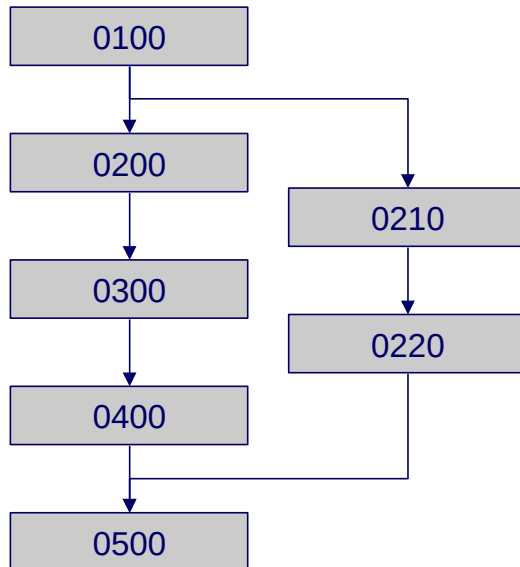
The standard sequence is available by default and describes the first sequence of the order.



If an order is sequentially processed, you only require the standard sequence. If you want to process specific operations in parallel or optionally to the standard sequence, you must create a new sequence including the respective operations. A standard sequence can then have parallel and alternative sequences. In this case, the standard sequence always has sequence number 0.

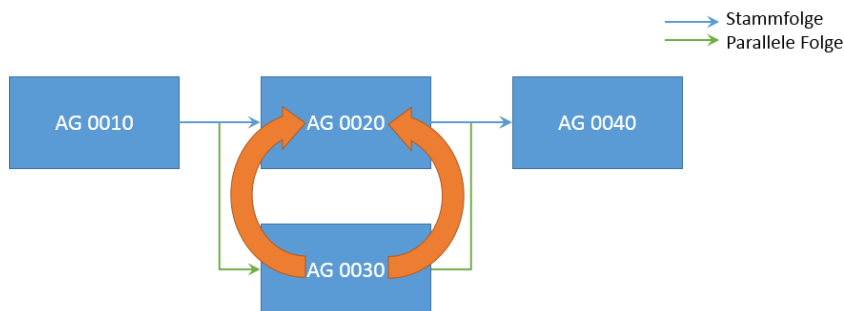
Parallel sequences

A parallel sequence runs in parallel to a partial sequence of the standard sequence. You use a parallel sequence, if specific processes must run at the same time. You use parallel sequences in process industry, for example.



In the example, the parallel sequence includes the operations 0210 and 0220. To position this parallel sequence, you must define an operation that identifies the start of the sequence (branch OP) and an operation that identifies the end of the sequence (return OP). You specify these operations in the application *Order sequences*.

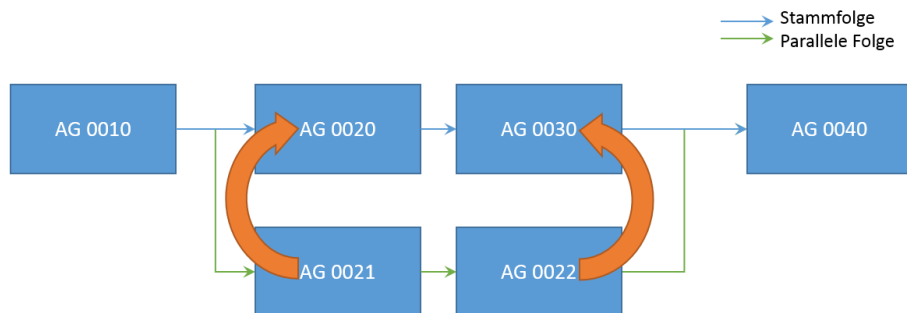
Example 1:



(blue = standard sequence; green = parallel sequence)

In the example, the parallel sequence only includes operation 0030. The respective branch OP for this parallel sequence is operation 0020, the return OP is also operation 0020.

Example 2:



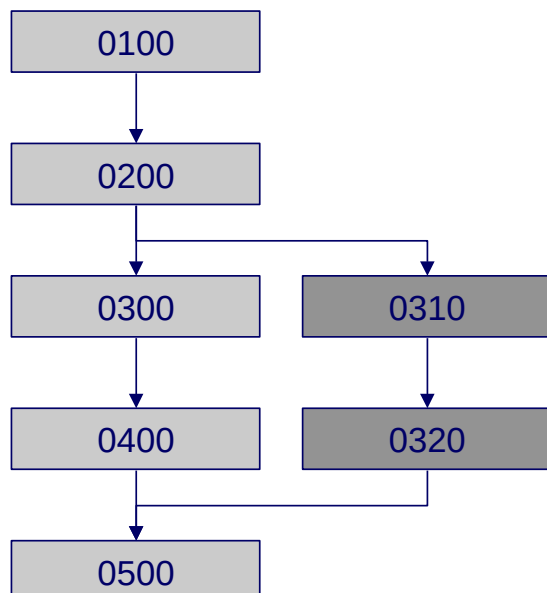
In the example, the parallel sequence includes the operation 0021 and the operation 0022. The respective branch OP for this parallel sequence is operation 0020, the return OP is operation 0030.

Alternative sequences

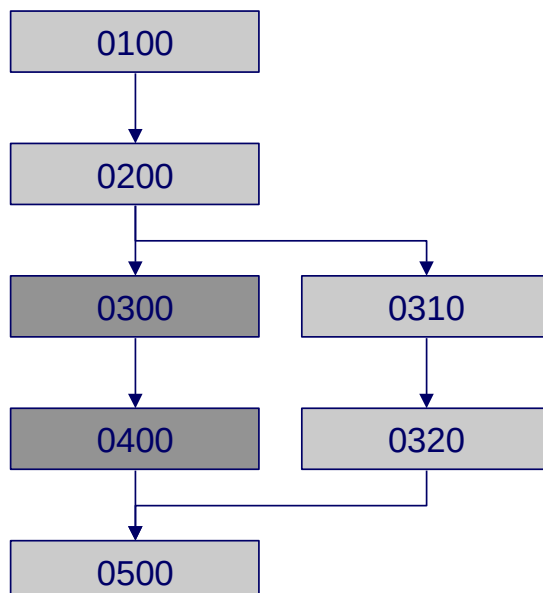
An alternative sequence describes one or more operations, which can be processed as an alternative to a partial sequence of the standard sequence. You use alternative sequences, if you have different production process for different batch sizes, for example.

If alternative sequences are available, only one of the alternative sequences is active and used for processing.

Order with an inactive alternative sequence



Order with an active alternative sequence

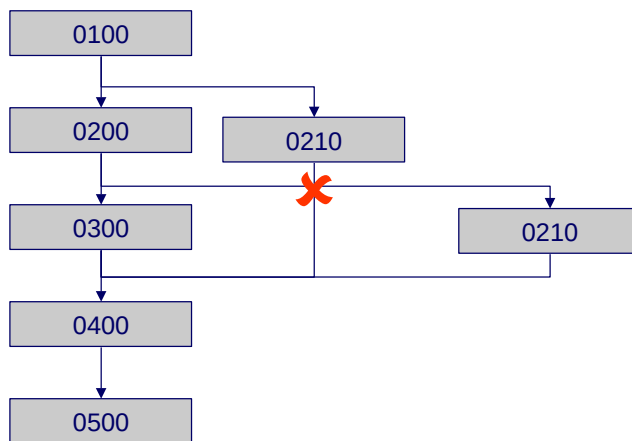


The system does not use inactive sequences. The inactive sequences are not used in scheduling or in detailed planning and you cannot make postings for inactive sequences.

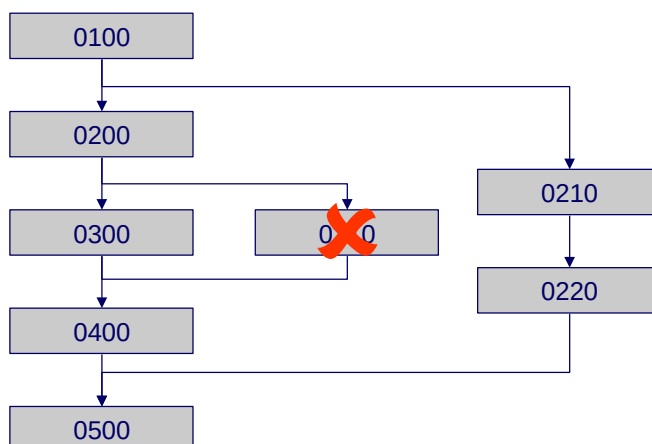
You can activate an alternative sequence on the client if specific conditions are fulfilled:

General

- Each operation is assigned to exactly one sequence.
- The system stores the resulting relationships between the operations and between the sequences in a table showing the internal relationships.
- Parallel sequences always run in parallel with the standard sequence.
- Alternative sequences are always an alternative to the standard sequence or to a part of the standard sequence.
- An overlapping of different operation sequences is not permitted.



- In case of a partial sequence of the standard sequence, which is specified using a branch OP and a return OP of a parallel or alternative sequence, another parallel or alternative sequence is not permitted that has a branch operation and/or return operation within the partial sequence.



Activating an alternative sequence

If you activate an alternative sequence, the respective part of the standard sequence is replaced by the alternative sequence. This partial sequence of the standard sequence is not run.

If the following conditions are fulfilled, you can NOT activate an alternative sequence:

- At least one operation of the partial sequence has already been started
- A second alternative sequence is already active in the partial sequence where this alternative sequence is included.

If you activate an alternative sequence, all other alternative sequences with the same branch and return OPs are deactivated.



If an alternative sequence is activated, the standard sequence is still identified as active.

Deactivating an alternative sequence

If you deactivate an alternative sequence, the respective part of the standard sequence is reactivated. This partial sequence is then run. If the following conditions are fulfilled, you can NOT deactivate an alternative sequence:

- At least one operation of the alternative sequence has already been started
- A second alternative sequence is already active in the partial sequence where this alternative sequence is included.

Deleting sequences

You can only delete a sequence that does not include any operations.

As a rule, you cannot delete a standard sequence.

Creating and deleting operations

If you create or delete an operation, the system automatically updates the order network for this order. The order network documents the relations between operations. The order network is used for the planning in the shop floor scheduling and for the processing/posting.

Copying an order

If an order is copied, the new order is available in its "initial state". This means that any existing alternative sequences are generally inactive, even if they were previously active in the order that was copied.

Operation status

Using the operation status, you cannot identify if the operation is part of an active alternative sequence or part of an inactive alternative sequence or part of an inactive partial sequence of the standard sequence (after activation of an alternative sequence). The operation status does not change during activation or deactivation.

Operations of active sequences have the initial status *prepared* when they are created. Operations of an inactive alternative sequence have the same status when they are created (*prepared*).

Sequencing list

The sequencing list does not display operations of an inactive alternative sequence or an inactive partial sequence of the standard sequence (after activation of an alternative sequence). You cannot log on these operations.

Merged operation

You cannot include operations in a merged operation that are part of an inactive alternative sequence or part of an inactive partial sequence of the standard sequence (after activation of an alternative sequence).

Specific features of parallel sequences

The planning algorithms in the shop floor scheduling also integrate parallel sequence structures.

If the option "Checking status of predecessor" is active and if you have combined several parallel sequences, you can only log on a succeeding operation when all sequences or more precisely the last operations of these sequences have been interrupted or finished. Any overlapping is identified.

To identify the send-ahead quantity of several parallel sequences, the system uses the smallest yield of the grouped sequences or of their last operations. The identified quantity is used as target quantity for the succeeding operations if the processing code of the respective operation includes a target quantity update.

Integration

To display operations of alternative sequences in the functions and evaluations on the client, note the following:

Operations / Operations logged on / Pool of orders

In the column *Control*, "Y" identifies an operation of an inactive sequence ("inactive operation").

If you do not want to display operations of inactive sequences in the order overview, then go to the selection panel, field *Control*: enable all options, but disable the option "Y".

Order overview

In tab *Progress*, the system displays operations of active and inactive sequences.

Order information

In the order information, the system displays operations of active and inactive sequences. If you want to identify the operations of inactive sequences, you must show the column *Control*. Use the column configurator of the operation table to show this column. In this column, the operations of inactive sequences have the identifier "Y".

Requirements

If you want to process sequences in the system, you require the respective license. You cannot use DOS based terminals (only applies when using HYDRA).

The following steps are required before use:

1. Configuration of the sequence number length in the system basic settings

WARNING

The sequence number length can only be configured during the initial system implementation process as long as no order backlog data is available in the system. A subsequent setting or change leads to inconsistent behavior.

2. Reactivation of the dynamic dialogs

As a result, the input fields on the Windows terminal will expand by the defined sequence number length (only applies when using HYDRA).

Selection criteria

The application provides the following selection criteria:

Order

Enter the number of the order to display the sequences of this order.

Field descriptions

Order number

Order that includes the specified sequence. You can only create a sequence for an order if the order header is already available in the system.

Sequence

Identification of the sequence within an order.

Note: The standard sequence always has the sequence number 0.



If the "sequence" field is not shown in the editing dialog, the sequence number length in the basic settings is 0. Please contact MPDV.

Name

Description of the sequence.

Sequence category

S = Standard sequence

For each order, there is exactly one standard sequence; you cannot delete the standard sequence.

P = Parallel sequence of the standard sequence

For each order, several parallel sequences can be available

A = Alternative sequence of the standard sequence

For each order, several alternative sequences can be available

Note:

After having created a sequence, you cannot change the category of the sequence.

Active

The identifier "Active" is only relevant for alternative sequences:

J = Active

N = Not active

If you create a new alternative sequence, this sequence is set to *not active*.

In case of standard sequences and alternative sequences, this identifier is always set to *Active*.

Alignment

If several parallel sequences are available, the sequences usually have a different lead time. If you select a specific sequence, you can have a resulting time buffer. The alignment function controls whether these buffers are at the beginning or the end of the sequence. The following variants are possible:

F = Earliest due date

If you use the earliest date to align the sequence, the buffer will be at the end of the sequence.

S = Latest due date

If you use the latest date to align the sequence, the buffer will be at the beginning of the sequence.

N = Irrelevant; this is the case for standard sequences and alternative sequences.

If several parallel sequences are available for a standard sequence, the system checks the alignment for all parts of the standard sequence that have a parallel sequence.

Version

Revision number/version, for information purposes only.

Branch operation

Number of an operation included in the standard sequence:

- parallel sequence: the parallel sequence starts *before* the specified operation
- alternative sequence: the operation of the alternative sequence starts instead of the specified operation.

In case of parallel and alternative sequences, this is a mandatory field. In case of a standard sequence, this field must be empty.

If you manually create an alternative or parallel sequence, the branch operation of the standard sequence must already be available in the pool of orders. If you transfer a sequence via interface, you must transfer a valid operation number for this field (no validation check is performed).

Return operation

Number of an operation included in the standard sequence:

- parallel sequence: the parallel sequence returns to the standard sequence *after* this OP.
- alternative sequence: the alternative sequence replaces the standard sequence *up to* this OP.

In case of parallel and alternative sequences, this is a mandatory field. In case of a standard sequence, this field must be empty.

If you manually create an alternative or parallel sequence, the return operation of the standard sequence must already be available in the pool of orders. If you transfer a sequence via interface, you must transfer a valid operation number for this field (no validation check is performed).

Reference sequence

The reference sequence specifies the sequence of the order that is identified via the two operations, branch OP and return OP. The reference sequence is *always* the standard sequence (sequence number 0).

In case of parallel and alternative sequences, this is a mandatory field. The standard sequence must already exist.

In case of a standard sequence, this field must be empty.

Toolbar



Activate

Activate an alternative sequence



Deactivate

Deactivate an alternative sequence



Edit orders

Calls the application [Edit orders](#).